GOWTHAM S
SRN: PES1UG24CS173
Section: C

LAB 7: FUZZ TESTING FOR INPUT VALIDATION BUGS

Hypothesis-Based Testing Report

1. Objective

To explore fuzz testing as a security and quality assurance technique by using the Hypothesis library to generate inputs, identify input-validation bugs in the provided Python application, and verify the corrected functions with property-based tests.

2. Testing Summary

The application contains three functions: **parse_plan**, **usage_average**, and **payable_bill**. Initial fuzz testing exposed failures when unexpected input values were supplied. Input validation was then added, and the updated test suite was run again.

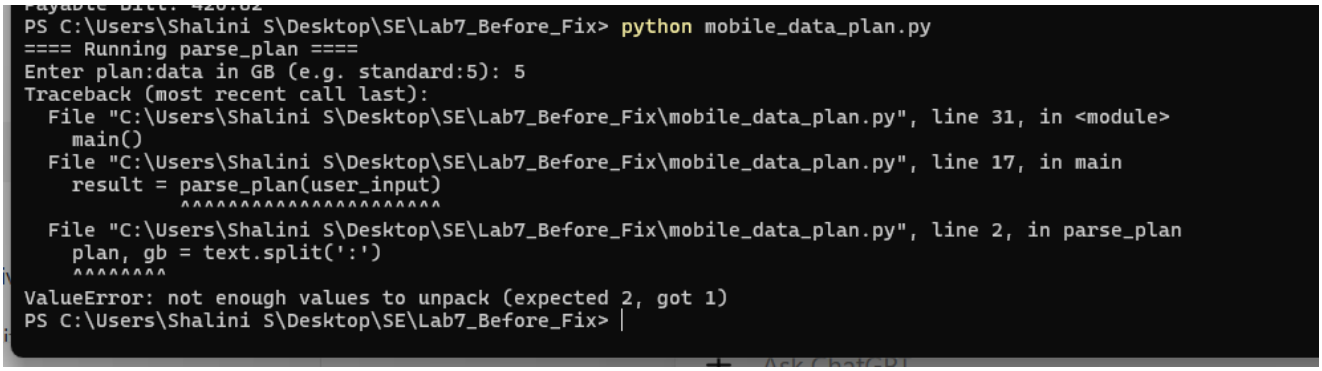
Initial result: 3 tests failed. **Final result:** 6 tests passed.

3. Screenshot Evidence

The screenshots are arranged chronologically, from the original application behaviour through initial fuzz testing to the final verified application.

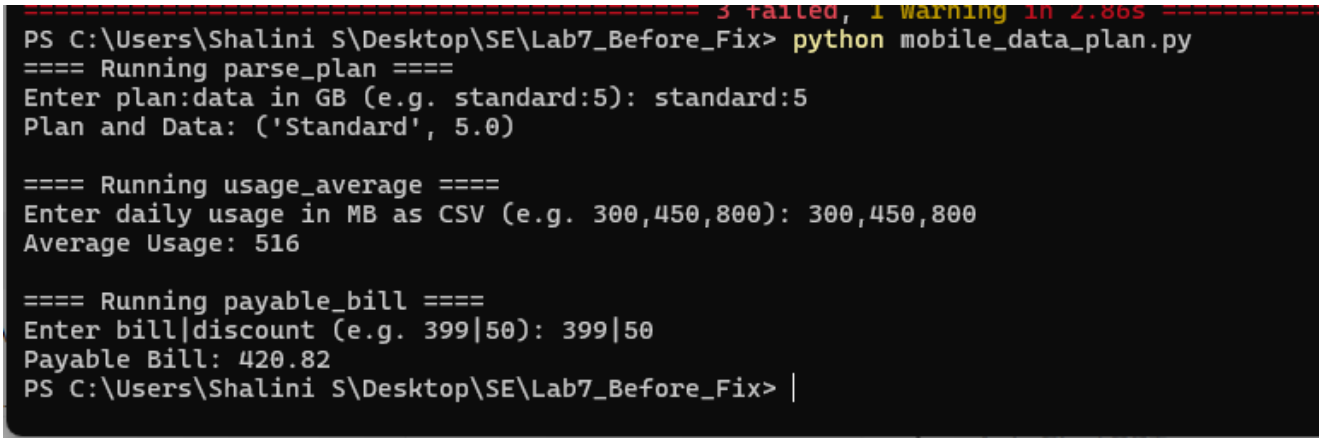
01. Original Application: Invalid Input Failure

The original application crashes when the plan input is entered without the required plan:data format. The traceback reports a ValueError because the input does not contain two parts.



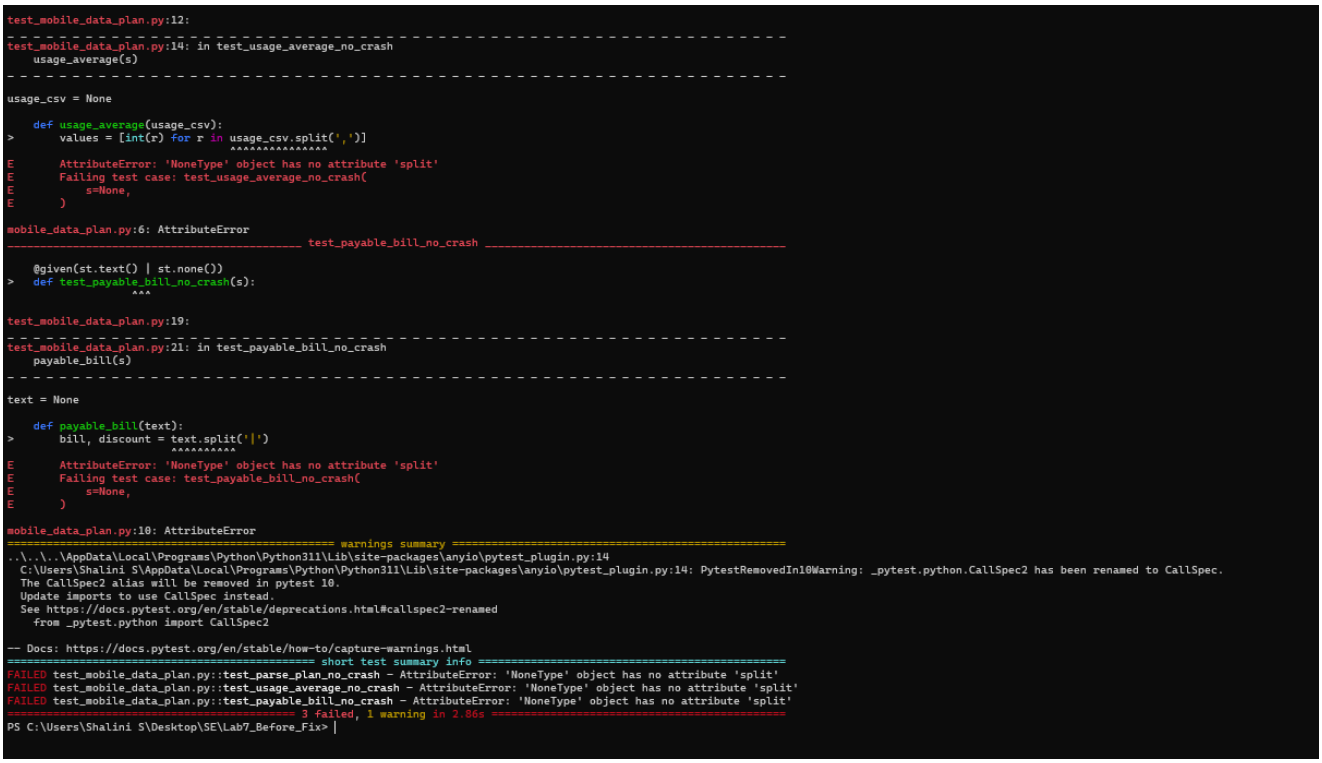
02. Original Application: Normal Execution

With correctly formatted inputs, the original application completes plan parsing, daily usage averaging, and payable bill calculation.



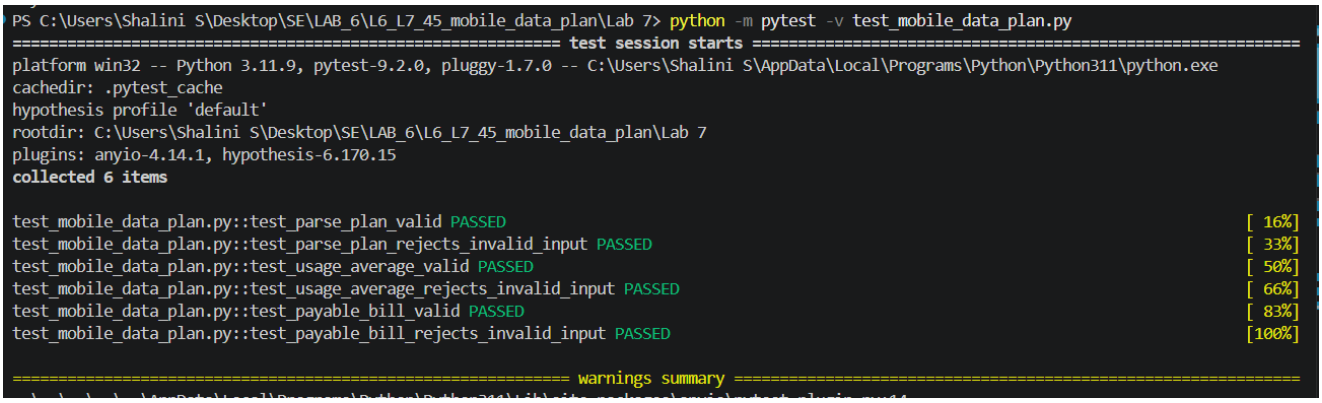
03. Initial Hypothesis Fuzz Testing: Failures Found

Initial fuzz testing exposes three failures. Hypothesis supplies None as input, causing AttributeError in parse_plan, usage_average, and payable_bill.



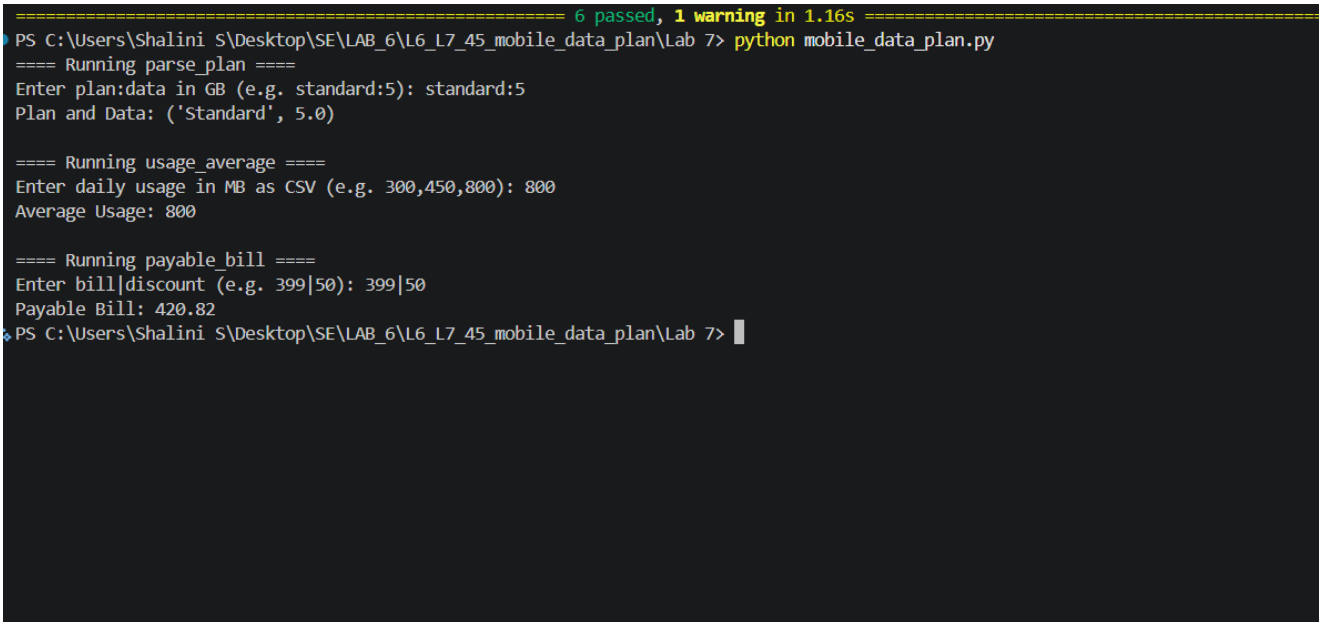
04. Updated Hypothesis Fuzz Testing: All Tests Passed

After input validation was added, all six property-based tests pass, covering valid and invalid inputs for the three functions.



05. Fixed Application: Successful Execution

The corrected application successfully processes the sample plan, usage, and bill inputs and displays the results.



4. Conclusion

Fuzz testing with Hypothesis helped reveal input-handling failures that were not visible during normal valid-input execution. After adding checks for input types, formats, numeric values, and invalid ranges, the updated test suite completed successfully with all six tests passing. The fixed application also completed its three sample operations successfully.

Final verification: 6 passed. The accompanying submission files are **mobile_data_plan.py** (fixed application) and **test_mobile_data_plan.py** (updated Hypothesis tests).

End of Report